

```

1  *-----
   -*
2  User:                gmark1
3  Date:                May 10, 2024
4  Time:                18:38:19
5  *-----
   -*
6  * Training Output
7  *-----
   -*
8
9
10
11
12 Variable Summary
13
14           Measurement      Frequency
15 Role           Level        Count
16
17 ID             INTERVAL      1
18 INPUT          INTERVAL      4
19 INPUT          NOMINAL       14
20
21
22
23 The CLUSTER Procedure
24 Average Linkage Cluster Analysis
25
26           Eigenvalues of the Covariance Matrix
27
28           Eigenvalue      Difference      Proportion      Cumulative
29
30      1      102.627890      65.089103          0.6844          0.6844
31      2       37.538788      29.703286          0.2504          0.9348
32      3        7.835502        7.408016          0.0523          0.9871
33      4         0.427485         0.140525          0.0029          0.9899

```

|    |    |          |          |        |        |
|----|----|----------|----------|--------|--------|
| 34 | 5  | 0.286960 | 0.047200 | 0.0019 | 0.9918 |
| 35 | 6  | 0.239760 | 0.032820 | 0.0016 | 0.9934 |
| 36 | 7  | 0.206940 | 0.012150 | 0.0014 | 0.9948 |
| 37 | 8  | 0.194790 | 0.041021 | 0.0013 | 0.9961 |
| 38 | 9  | 0.153769 | 0.032318 | 0.0010 | 0.9971 |
| 39 | 10 | 0.121451 | 0.033859 | 0.0008 | 0.9979 |
| 40 | 11 | 0.087592 | 0.026660 | 0.0006 | 0.9985 |
| 41 | 12 | 0.060932 | 0.009146 | 0.0004 | 0.9989 |
| 42 | 13 | 0.051786 | 0.018163 | 0.0003 | 0.9993 |
| 43 | 14 | 0.033624 | 0.013758 | 0.0002 | 0.9995 |
| 44 | 15 | 0.019865 | 0.001753 | 0.0001 | 0.9996 |
| 45 | 16 | 0.018112 | 0.008381 | 0.0001 | 0.9998 |
| 46 | 17 | 0.009731 | 0.002631 | 0.0001 | 0.9998 |
| 47 | 18 | 0.007100 | 0.002430 | 0.0000 | 0.9999 |
| 48 | 19 | 0.004670 | 0.001581 | 0.0000 | 0.9999 |
| 49 | 20 | 0.003090 | 0.000230 | 0.0000 | 0.9999 |
| 50 | 21 | 0.002860 | 0.001122 | 0.0000 | 0.9999 |
| 51 | 22 | 0.001737 | 0.000271 | 0.0000 | 0.9999 |
| 52 | 23 | 0.001467 | 0.000345 | 0.0000 | 1.0000 |
| 53 | 24 | 0.001122 | 0.000154 | 0.0000 | 1.0000 |
| 54 | 25 | 0.000968 | 0.000144 | 0.0000 | 1.0000 |
| 55 | 26 | 0.000824 | 0.000116 | 0.0000 | 1.0000 |
| 56 | 27 | 0.000708 | 0.000130 | 0.0000 | 1.0000 |
| 57 | 28 | 0.000579 | 0.000036 | 0.0000 | 1.0000 |
| 58 | 29 | 0.000542 | 0.000112 | 0.0000 | 1.0000 |
| 59 | 30 | 0.000430 | 0.000036 | 0.0000 | 1.0000 |
| 60 | 31 | 0.000395 | 0.000086 | 0.0000 | 1.0000 |
| 61 | 32 | 0.000308 | 0.000092 | 0.0000 | 1.0000 |
| 62 | 33 | 0.000216 | 0.000116 | 0.0000 | 1.0000 |
| 63 | 34 | 0.000100 | 0.000012 | 0.0000 | 1.0000 |
| 64 | 35 | 0.000088 | 0.000015 | 0.0000 | 1.0000 |
| 65 | 36 | 0.000073 | 0.000025 | 0.0000 | 1.0000 |
| 66 | 37 | 0.000048 | 0.000048 | 0.0000 | 1.0000 |
| 67 | 38 | 0.000000 | 0.000000 | 0.0000 | 1.0000 |
| 68 | 39 | 0.000000 | 0.000000 | 0.0000 | 1.0000 |
| 69 | 40 | 0.000000 | 0.000000 | 0.0000 | 1.0000 |

|    |    |           |          |         |        |
|----|----|-----------|----------|---------|--------|
| 70 | 41 | 0.000000  | 0.000000 | 0.0000  | 1.0000 |
| 71 | 42 | 0.000000  | 0.000000 | 0.0000  | 1.0000 |
| 72 | 43 | 0.000000  | 0.000000 | 0.0000  | 1.0000 |
| 73 | 44 | 0.000000  | 0.000000 | 0.0000  | 1.0000 |
| 74 | 45 | -0.000000 | 0.000000 | -0.0000 | 1.0000 |
| 75 | 46 | -0.000000 | 0.000000 | -0.0000 | 1.0000 |
| 76 | 47 | -0.000000 | 0.000000 | -0.0000 | 1.0000 |
| 77 | 48 | -0.000000 | 0.000000 | -0.0000 | 1.0000 |
| 78 | 49 | -0.000000 | 0.000000 | -0.0000 | 1.0000 |
| 79 | 50 | -0.000000 | 0.000000 | -0.0000 | 1.0000 |
| 80 | 51 | -0.000000 |          | -0.0000 | 1.0000 |

81

82 Root-Mean-Square Total-Sample Standard Deviation 1.768077

83

84 Root-Mean-Square Distance Between Observations 17.8567

85

86

87 Cluster History

88 Number

89 of

Ps

|    |          |                           |          |      |     |
|----|----------|---------------------------|----------|------|-----|
|    | eudo F   | Pseudo                    | Norm RMS |      |     |
| 90 | Clusters | -----Clusters Joined----- |          | Freq | Sta |
|    | tistic   | t-Squared                 | Distance | Tie  |     |

91

|    |    |      |      |     |  |
|----|----|------|------|-----|--|
| 92 | 49 | OB22 | OB42 | 373 |  |
|----|----|------|------|-----|--|

|     |     |        |  |  |
|-----|-----|--------|--|--|
| 943 | 6.5 | 0.2315 |  |  |
|-----|-----|--------|--|--|

|    |    |      |      |     |  |
|----|----|------|------|-----|--|
| 93 | 48 | OB16 | OB35 | 562 |  |
|----|----|------|------|-----|--|

|     |      |        |  |  |
|-----|------|--------|--|--|
| 946 | 66.1 | 0.2393 |  |  |
|-----|------|--------|--|--|

|    |    |      |      |     |  |
|----|----|------|------|-----|--|
| 94 | 47 | CL48 | CL49 | 935 |  |
|----|----|------|------|-----|--|

|     |     |       |  |  |
|-----|-----|-------|--|--|
| 932 | 127 | 0.247 |  |  |
|-----|-----|-------|--|--|

|    |    |      |      |      |  |
|----|----|------|------|------|--|
| 95 | 46 | CL47 | OB45 | 1095 |  |
|----|----|------|------|------|--|

|     |      |       |  |  |
|-----|------|-------|--|--|
| 939 | 49.1 | 0.254 |  |  |
|-----|------|-------|--|--|

|    |    |      |      |      |  |
|----|----|------|------|------|--|
| 96 | 45 | CL46 | OB39 | 1170 |  |
|----|----|------|------|------|--|

|     |      |        |  |  |
|-----|------|--------|--|--|
| 953 | 24.5 | 0.2565 |  |  |
|-----|------|--------|--|--|

|    |    |      |      |      |  |
|----|----|------|------|------|--|
| 97 | 44 | CL45 | OB30 | 1326 |  |
|----|----|------|------|------|--|

|     |      |        |  |  |
|-----|------|--------|--|--|
| 949 | 85.3 | 0.2647 |  |  |
|-----|------|--------|--|--|

|     |      |      |        |      |
|-----|------|------|--------|------|
| 98  | 43   | CL44 | OB37   | 1462 |
|     | 953  | 56.6 | 0.2777 |      |
| 99  | 42   | OB13 | CL43   | 1485 |
|     | 973  | 11.9 | 0.2836 |      |
| 100 | 41   | OB28 | OB48   | 87   |
|     | 996  | 2.7  | 0.2856 |      |
| 101 | 40   | OB31 | OB41   | 146  |
|     | 1018 | 11.9 | 0.2857 |      |
| 102 | 39   | OB17 | CL41   | 184  |
|     | 1036 | 23.4 | 0.292  |      |
| 103 | 38   | OB8  | OB43   | 220  |
|     | 1048 | 45.4 | 0.2997 |      |
| 104 | 37   | CL42 | CL39   | 1669 |
|     | 1038 | 109  | 0.3029 |      |
| 105 | 36   | OB23 | CL40   | 173  |
|     | 1062 | 15.5 | 0.3091 |      |
| 106 | 35   | OB19 | CL36   | 196  |
|     | 1089 | 9.0  | 0.315  |      |
| 107 | 34   | CL38 | OB25   | 227  |
|     | 1120 | 5.2  | 0.3172 |      |
| 108 | 33   | OB7  | OB24   | 175  |
|     | 1139 | 42.2 | 0.3173 |      |
| 109 | 32   | OB4  | OB47   | 106  |
|     | 1166 | 23.3 | 0.3184 |      |
| 110 | 31   | CL37 | OB49   | 1675 |
|     | 1203 | 5.5  | 0.3184 |      |
| 111 | 30   | CL31 | OB18   | 1692 |
|     | 1236 | 22.4 | 0.3389 |      |
| 112 | 29   | CL30 | CL35   | 1888 |
|     | 1179 | 239  | 0.355  |      |
| 113 | 28   | OB33 | OB44   | 101  |
|     | 1216 | 14.4 | 0.3607 |      |
| 114 | 27   | CL33 | OB14   | 183  |
|     | 1259 | 10.2 | 0.3707 |      |
| 115 | 26   | CL29 | OB32   | 1900 |
|     | 1299 | 20.8 | 0.3744 |      |

|     |      |      |        |      |
|-----|------|------|--------|------|
| 116 | 25   | OB3  | OB40   | 122  |
|     | 1333 | 39.1 | 0.3744 |      |
| 117 | 24   | OB21 | CL28   | 106  |
|     | 1387 | 8.0  | 0.3856 |      |
| 118 | 23   | CL34 | CL24   | 333  |
|     | 1376 | 143  | 0.4122 |      |
| 119 | 22   | OB11 | OB12   | 19   |
|     | 1440 | 4.6  | 0.4166 |      |
| 120 | 21   | OB2  | OB5    | 66   |
|     | 1496 | 33.9 | 0.4172 |      |
| 121 | 20   | CL32 | OB9    | 107  |
|     | 1573 | 3.1  | 0.4217 |      |
| 122 | 19   | OB1  | OB46   | 22   |
|     | 1653 | 12.9 | 0.4522 |      |
| 123 | 18   | OB20 | OB36   | 12   |
|     | 1745 | 11.9 | 0.4568 |      |
| 124 | 17   | OB15 | OB38   | 61   |
|     | 1845 | 15.7 | 0.4646 |      |
| 125 | 16   | CL25 | OB10   | 190  |
|     | 1883 | 107  | 0.4657 |      |
| 126 | 15   | CL26 | OB29   | 1903 |
|     | 2011 | 11.2 | 0.474  |      |
| 127 | 14   | CL21 | CL27   | 249  |
|     | 2058 | 135  | 0.482  |      |
| 128 | 13   | CL20 | OB50   | 111  |
|     | 2218 | 17.6 | 0.5214 |      |
| 129 | 12   | CL14 | CL17   | 310  |
|     | 2284 | 111  | 0.5285 |      |
| 130 | 11   | CL18 | OB26   | 13   |
|     | 2509 | 3.6  | 0.5465 |      |
| 131 | 10   | CL23 | CL15   | 2236 |
|     | 1925 | 1323 | 0.5532 |      |
| 132 | 9    | CL19 | CL16   | 212  |
|     | 2124 | 45.6 | 0.5741 |      |
| 133 | 8    | CL9  | OB6    | 224  |
|     | 2389 | 33.0 | 0.6366 |      |

|     |      |      |        |      |
|-----|------|------|--------|------|
| 134 | 7    | CL8  | CL11   | 237  |
|     | 2738 | 32.3 | 0.6686 |      |
| 135 | 6    | CL13 | OB27   | 114  |
|     | 3267 | 22.5 | 0.6827 |      |
| 136 | 5    | CL7  | CL22   | 256  |
|     | 3948 | 57.3 | 0.7305 |      |
| 137 | 4    | CL5  | OB34   | 257  |
|     | 5235 | 8.1  | 1.1189 |      |
| 138 | 3    | CL4  | CL6    | 371  |
|     | 5059 | 841  | 1.2    |      |
| 139 | 2    | CL12 | CL10   | 2546 |
|     | 2543 | 6220 | 1.3513 |      |
| 140 | 1    | CL3  | CL2    | 2917 |
|     | .    | 2543 | 1.6516 |      |

141

142

143

144 Candidates for Optimum Number of Clusters

145

|     |     |          |            |
|-----|-----|----------|------------|
| 146 |     | Number   | Clustering |
| 147 |     | of       | Cubic      |
| 148 | Obs | Clusters | Criterion  |

149

|     |    |    |         |
|-----|----|----|---------|
| 150 | 1  | 4  | 58.7451 |
| 151 | 2  | 8  | 32.9243 |
| 152 | 3  | 11 | 49.6157 |
| 153 | 4  | 13 | 47.2600 |
| 154 | 5  | 15 | 46.0131 |
| 155 | 6  | 17 | 45.0177 |
| 156 | 7  | 24 | 39.4709 |
| 157 | 8  | 30 | 41.3639 |
| 158 | 9  | 34 | 40.2084 |
| 159 | 10 | 39 | 41.1504 |

160

161

162 \*-----

```

- *
163 * Score Output
164 *-----
- *

165
166
167 *-----
- *

168 * Report Output
169 *-----
- *

170
171
172
173
174 Variable Importance
175
176                                     Number of      Number
      of                                     Splitting    Surrog
177                                     ate
178 Variable Name      Label      Rules      Rule
      s      Importance
179
180 BMI                BMI                0                10
      1.00000
181 General_Health                0                11
      0.94290
182 Smoking            Smoking            1                0
      0.74254
183 PhysicalHealth      PhysicalHealth      5                3
      0.72773
184 SleepTime            SleepTime            0                8
      0.68321
185 Difficult_Walking                2                0
      0.68069

```

|     |                  |                  |   |   |
|-----|------------------|------------------|---|---|
| 186 | AgeCategory      | AgeCategory      | 0 | 4 |
|     | 0.66817          |                  |   |   |
| 187 | HeartDisease     | HeartDisease     | 1 | 2 |
|     | 0.66442          |                  |   |   |
| 188 | MentalHealth     | MentalHealth     | 1 | 8 |
|     | 0.66375          |                  |   |   |
| 189 | KidneyDisease    | KidneyDisease    | 1 | 0 |
|     | 0.62816          |                  |   |   |
| 190 | Race             | Race             | 0 | 2 |
|     | 0.61648          |                  |   |   |
| 191 | PhysicalActivity | PhysicalActivity | 2 | 5 |
|     | 0.56772          |                  |   |   |
| 192 | Sex              | Sex              | 3 | 0 |
|     | 0.48180          |                  |   |   |
| 193 | Asthma           | Asthma           | 1 | 2 |
|     | 0.36443          |                  |   |   |
| 194 | Diabetic         | Diabetic         | 0 | 4 |
|     | 0.30923          |                  |   |   |
| 195 | AlcoholDrinking  | AlcoholDrinking  | 2 | 0 |
|     | 0.26167          |                  |   |   |